ANOVA & Sum of Squares

Based off of 9ST0 Content
lbweb.uk/publications

Analysis of Variance, or ANOVA, is to test means across three or more groups to see if at least one of them is different to the rest of the samples.
This is done by looking at the Spread (variance) between the samples rather than within the samples.
If the variation between groups is massively larger than the random noise inside them, you conclude the means are not all equal.

Blocking factors are when you intentionally group data together to isolate and remove variation caused by external factors.

- e.g. You have a sample of 12 plants however 5 used a seperate and less expensive fertiliser. This is therefore likely to cause variance in the sample, so we group each plant by the fertiliser used. Another blocking factor in the same sample, blocking factor B, may be the location they grew in a warehouse.
- Adding these blocking factors mean you get a smaller residual/error

Residual / Error is the representation of unaccounted for unpredictable real-world variation that your model cannot explain. It's whatever is left after we calculate how much variation is caused by either blocking factor

Sum of Squares is the measure of the total amount of variation in your data.

Degrees of freedom is the representation of how many independant values are able to vary once the mean is calculated.

- e.g. if you have a sample size of 5 with mean 10, $df = 5-1 = 4$.
  Therefore 4 values are able to vary from the mean thus creating variation, and that 1 value must stay the same to ensure the mean stays at 10.

Mean of Squares is the average variance per degree of freedom.

F-number is your test statistic.

Required Formulae:

- Sum of Squares (SS) → $SS_T = \sum x_{ij}^2 - \frac{(\sum x)^2}{n}$
  - Where x_{ij} represents $x_{row,\ column}$
    - e.g. $x_{2,3}$ represents the value in row 2 column 3
- Sum of Squares for a blocking factor → $SS_A = \sum \frac{T_i}{n_i} - \frac{(\sum x)^2}{n}$
  - Where n_i represents the number of values in a row or column
  - Where T_i is the row or column's total
- Residual / Error Sum of Squares → $SS_{R/E} = SS_T - (SS_A + SS_B)$
- Degrees of freedom (df) → $df = n_i - 1$
  - Where n_i is either the number of values in a row or column, or total number of values.
- Mean of Squares (MS) → $MS = SS \div df$
- F number → $F = MS \div MS_R$
  - Where MS_R is the mean of Squares of the residual / error band.
- As shown by $SS_{R/E}$, $MS_{R/E}$ and $df_{R/E}$ is calculated the same way just with df or ms instead of ss

Example

Example Data		Camper		
		Olan	Majid	Ralphet
Model	A	42	29	19
	B	37	33	24
	C	24	29	18
	D	25	22	13

An outdoor equipment researcher investigates whether the sleep score achieved by campers depends on the tent model used. Four tent models (A, B, C, and D) are tested. Three campers (Olan, Majid, and Ralphet) act as blocking factors, with each camper testing all four tent models in randomized trials.

The resulting sleep scores (out of 50) are recorded in the table

Statistics Mode on the CG50 where data is all under List 1 and frequency is 1

$\Sigma x = 315$

$\Sigma x^2 = 9019$

$n = 12$

To calculate the "ANOVA Table", either use Stats mode or formulae below:

$$SS_{Total} = 9019 - \frac{315^2}{12} = 750.25$$

Example Data		Camper			T
		Olan	Majid	Ralphet	
Model	A	42	29	19	90
	B	37	33	24	94
	C	24	29	18	71
	D	25	22	13	60
	T	128	113	74	315

Camper is blocking factor A

Model is blocking factor B

Remember, $SS_A = \Sigma \frac{T_i}{n_i} - \frac{(\Sigma x)^2}{n}$

$$\therefore SS_A = \left(\frac{128^2}{4} + \frac{113^2}{4} + \frac{74^2}{4}\right) - \frac{315^2}{12} = 388.5$$

→ Since each Camper has 4 values

$$\therefore SS_B = \left(\frac{90^2}{3} + \frac{94^2}{3} + \frac{71^2}{3} + \frac{60^2}{3}\right) - \frac{315^2}{12} = 256.92$$

→ Since each Model has 3 values

Remember, $SS_{R/E} = SS_T - (SS_A + SS_B)$

$\therefore SS_{R/E} = 750.25 - (388.5 + 256.92) = 104.83$

ANOVA Test, One-tailed at 5% Significance.

Factor A =: H_0: There is no difference in mean Sleep Scores between the Campers

H_1: There is a difference in mean Sleep Scores between the Campers.

Factor B = H_0: There is no difference in mean Sleep Score between the four tent models

H_1: There is a difference in mean Sleep Score between the four tent models

Remember, $df = n - 1$

$\therefore df_A = 3 - 1 = 2$ → Since 3 Campers

$\therefore df_B = 4 - 1 = 3$ → Since 4 models

$\therefore df_T = 12 - 1 = 11$

$\therefore df_{R/E} = 11 - (3 + 2) = 6$

Remember, $MS = SS \div df$

$\therefore MS_A = \frac{388.5}{2} = 194.25$

$\therefore MS_B = \frac{256.92}{3} = 85.64$

$\therefore MS_{R/E} = \frac{104.83}{6} = 17.47$

Remember, $F = MS \div MS_{R/E}$

$\therefore F_A = \frac{194.25}{17.47} = 11.12$

$\therefore F_B = \frac{85.64}{17.47} = 4.902$

Finally giving us the following:

Results/Answers	SS	DF	MS	F
BF. A	388.5	2	194.25	11.12
BF. B	256.92	3	85.64	4.902
Error	104.83	6	17.47	—
Total	750.25	11	—	—

Comparing against the CV:

Databooklet Page 21 & 22. Use the table with your α
Here, we use the table where "$(p = 0.95)$" since our $\alpha = 1 - 0.05 = 0.95$
V_1 is your blocking factors df and V_2 is your error's df
Then compare both blocking factors to their respective CV.

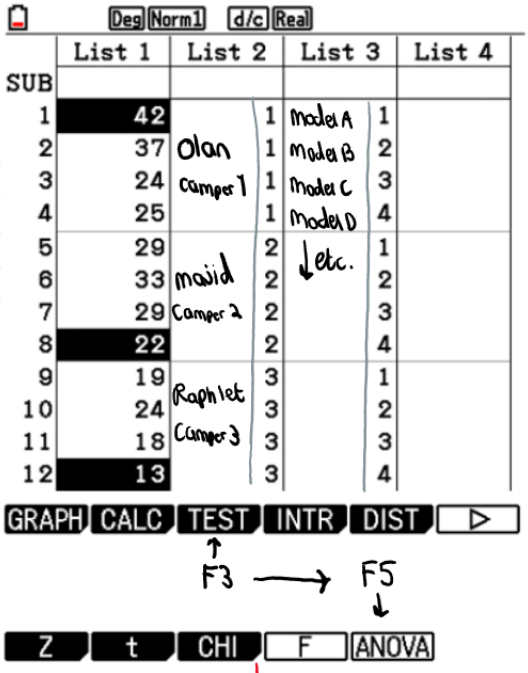
F value ↓

CV_A: $V_1 = 11$ & $V_2 = 2$ $\therefore CV_A = 3.982$ $\therefore 11.12 > 3.982$ $\therefore$ Fail to accept H_0
CV_B: $V_1 = 11$ & $V_2 = 3$ $\therefore CV_B = 3.587$ $\therefore 4.902 > 3.587$ $\therefore$ Fail to accept H_0

If using the CG50:

List 1 is your data going like this:

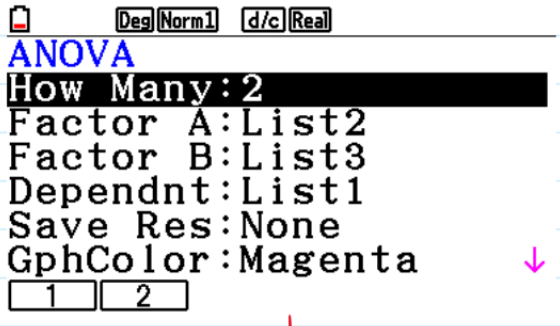
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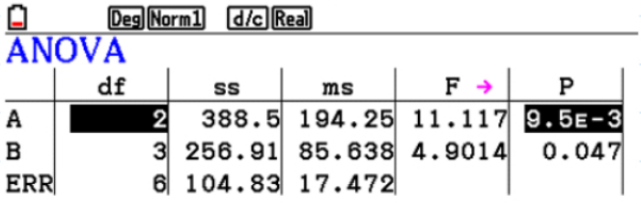


List 2 is the position the values in within blocking factor A

List 3 is the position the values in within blocking factor B.

If you have only one blocking factor, Set "how many?" to 1 and leave list 3 (blocking factor B) blank.





p is the probability of the F-value occurring if we accept H_0
↓ Same as Chi-Squared.

ERR is error or residual.
A and B are the blocking factors.